

ANNEX D – CORE PIPELINE SPECIFICATION FOR S2S SYSTEM

Filed as supporting documentation for the specification titled:

“Stardust to Sovereignty: Resonance-Based Knowledge Organization System.”

FIELD OF THE INVENTION

The invention relates to computer-implemented data-processing systems and methods for measuring and organizing information through resonance-based coherence computation.

BACKGROUND

Conventional data-organization systems rely on optimization and probability models that fail to detect intrinsic alignment or coherence across datasets. Such systems often yield unstable results or require manual post-processing to identify meaningful correlations. The disclosed S2S pipeline introduces a resonance-domain architecture that automatically detects and stabilizes coherence states within digital data, providing improved computational efficiency and interpretability.

SUMMARY

The S2S Core Pipeline implements a complete, computer-executed workflow that transforms unstructured digital data into coherence-ordered structures. The process includes five primary operations—parsing, computing resonance, updating coherence, detecting halting, and translating output. Each operation is performed by a processor or distributed computing node executing stored instructions on a non-transitory computer-readable medium. The system provides measurable transformation of data, verified through reproducible resonance and convergence metrics.

DETAILED DESCRIPTION OF THE INVENTION

1. PIPELINE OVERVIEW

The S2S Core Pipeline provides an end-to-end computational process executed by one or more processors. Input data are parsed into resonance domains; inter-domain relationships are analyzed to compute resonance metrics (R_{ij}); global coherence is monitored through a derivative function (dC/dt); and stable resonance matrices are translated into symbolic output. See FIG. 1 for the general system architecture.

2. CORE PIPELINE STEPS

Parse Content – The system parses unstructured digital input into resonance domains using natural-language tokenization, keyword extraction, and domain-association mapping (FIG. 1).

Compute Resonance – A processor executes similarity algorithms to compute R_{ij} values between resonance domains using cosine similarity, Jaccard overlap, and temporal-decay weighting (FIG. 2).

Update Graph – The system filters resonance values above a threshold ($R > 0.5$) and updates the global coherence matrix Ψ accordingly.

Detect Halting – A halting-factor module monitors the derivative of global coherence (dC/dt) and terminates computation when $|dC/dt| < \epsilon$, typically $\epsilon = 0.001$ (FIG. 3).

Translate Output – Stabilized coherence matrices Ψ are passed to a translation operator $T(\Psi) \rightarrow S$ that produces symbolic or visual representations for user interpretation (FIG. 5).

Each step is implemented in software and may execute on CPU, GPU, or distributed architectures connected via a data-exchange protocol.

3. MATHEMATICAL FOUNDATION

Resonance between domains i and j is computed as $R_{ij} = (w_1 \times \text{vectorSimilarity} + w_2 \times \text{domainOverlap} + w_3 \times \text{temporalDecay})$, where $w_1 = 0.4$, $w_2 = 0.4$, and $w_3 = 0.2$.

Global coherence is monitored as $C(t) = \sum(R_{ij}) / n^2$.

Convergence occurs when $|dC/dt| < 0.001$, indicating stable coherence.

These quantitative operations convert unstructured digital input into measurable coherence states, representing a practical application of mathematical relationships within a computing environment.

4. IMPLEMENTATION FEATURES

The system executes on standard computing hardware and may be deployed as local software, distributed cloud processes, or embedded firmware.

It supports runtime error handling, timing measurement, and resource optimization.

Each module can be independently configured and tested to ensure deterministic outcomes. (See FIG. 4 for data-flow schematic of Psi propagation.)

5. EXAMPLE EXECUTION

In one embodiment, three text inputs were processed through the pipeline.

The system produced $R_{ij} = 0.561353$, detected halting at iteration 13 when $dC/dt = 0.000460$, and generated a symbolic output identifying two aligned resonance domains. The method achieves a concrete data transformation from unstructured input to stable coherence matrices, enhancing computational reliability and interpretability.

CLOSING HARDWARE STATEMENT

The disclosed method may be embodied in hardware, firmware, software, or any combination thereof and executed on one or more processors connected by a computing network. The process may also be implemented on GPU-accelerated matrix-computation hardware for high-throughput resonance evaluation.

CLAIMS (PLACEHOLDER)

No claims are presented in this provisional application. In one possible embodiment, the claimed invention may include a computer-implemented method comprising the steps described above.

ABSTRACT

A computer-implemented method for organizing information through resonance-based coherence computation, comprising: parsing digital input data; computing resonance metrics between resonance domains; detecting convergence using halting conditions; and translating coherent states into symbolic outputs. The method transforms unstructured data into coherence-ordered structures and provides a practical improvement in computational stability and efficiency.

TECHNICAL PROBLEM / SOLUTION

Problem: Conventional AI and data-organization systems rely on probability and optimization but cannot detect intrinsic alignment or coherence in data, leading to instability and loss of semantic integrity.

Solution: The S2S Resonance Engine computes quantitative coherence metrics and halts processing when stability is achieved, transforming unstructured information into coherence-ordered representations and thereby improving computer functionality.